

FIN294 R&D Quintiles Project

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EXECUTIVE MEMO

Updated to include reconciliation results as of 2/22/2026

In this project, we were tasked with testing and evaluating the idea of Research and Development costs as a portfolio indicator for alpha generation. This project was taken on in many steps, from a simple binary portfolio all the way to quintile portfolio generation and long-short portfolio analysis. In the end, we found that by using long-short portfolios created from the highest and lowest quintiles of R&D generation, we were able to find appreciable alphas.

The first step of the project involved creating binary portfolios, one with firms with no R&D and one with firms with any positive R&D. While conceptually simple, each person approached the details of the project differently, and once we had all pulled and merged the data multiple rounds of data policy affirming and reconciliation were needed. We learned how much effort must go into creating an airtight data policy, and how even with a unified policy some differences in results will remain. Once these rounds of reconciliation were completed, we tackled the final step of the project: creating quintile portfolios and running regressions across both these portfolios and the long-short portfolios comparing the highest R&D quintile to both no R&D and the lowest R&D quintile.

Our first decision to be made regarding these quintiles was how to treat the R&D variable. Where our binary portfolio only cared about whether there was a signal, the quintile cared about strength of signal. To account for the lasting effects of R&D across multiple years, we decided to use a weighted R&D measure. Using the formula

$$XRD_{Total} = XRD_t + 0.8XRD_{t-1} + 0.6XRD_{t-2} + 0.4XRD_{t-3} + 0.2XRD_{t-4}$$

we took into account these effects. However, on our initial passthrough we did not weigh this signal based on the market cap of each company. This led to our weighted XRD variable becoming somewhat of a proxy for the size measure, with larger companies more likely to have high XRD values and smaller companies likely to have smaller ones. On revision, we took this weighted XRD variable and divided it by the market cap for the company from the previous month. This allowed the variables to be scaled as a percentage of company size, meaning that we did not favor large companies in the way that the initial variable did. Once this change was made, we still see the size factor as the most significant factor, but the size differences between portfolios are less extreme than before we normalized the variables.

With this variable decided, we then cut the portfolio into 5 portfolios with 20% of the total companies in each quintile. We also created two more portfolios, a long-short portfolio subtracting Q1 (lowest R&D) from Q5 (highest R&D), and a long-short portfolio subtracting the no R&D portfolio from Q5. Each of these portfolios was then regressed using both the CAPM and the Fama-French 3 Factor model as the benchmark. They were also checked across two different time periods: from 1980-2000 and from 2001-2022, and both equal-weighting and value-weighting were used. Finally, we used one sample that had all firms in it, and one that excluded any firms with a market cap under \$100 million as well as any firms in the top 100 in market cap. See appendix A of this document for the results from the regressions of the long-short portfolios.

In these portfolios, we see significant alphas in the following conditions: 1980-2000 equal-weighted portfolios with all firms, Q5-Q1 equal-weighted portfolios from 2001-2022, Q5-Q1 equal-weighted portfolios from 1980-2000 excluded firms with market caps less than \$100 million, and the Q5-Q0 Value Weighted Fama-French portfolio from 1980-2000 excluding small cap firms. In all cases, XRD seems to be a stronger indicator in the 1980-2000 range than it does in the 2001-2022 range. Seeing most of the alphas in equal-weighted portfolios seems to indicate that this effect is driven by small cap stocks, as it is seen more strongly when all companies regardless of size are given equal weight. Larger firms are also less likely to experience mispricing, so giving equal weight to all firms gives more weight to stocks that have a higher likelihood of being mispriced. In a similar vein to this, as time has progressed, we have become better at pricing intangible assets such as R&D. Reg FD, which came out in 2000, improved how R&D values were reported, so the benefits of this R&D expenditure are more likely to be baked into the prices of stocks.

In the end, our group concludes that while R&D intensity could have been a useful indicator for portfolio analysis in the past, signs point to this variable becoming a less lucrative portfolio indicator as time goes on. Future endeavors could involve using sector-specific universes to analyze R&D impact on a distinct subset of stocks, or choosing any number of different variables to create new portfolios.

Post Reconciliation Update

In reconciling with the other groups in the class, we agreed on a few data policies to help align our data. First, we decided to use the SQL queries to WRDS from one of the groups to ensure that we all started with the exact same dataset. Next, we agreed to weight our XRD variable using only the market cap of the current year instead of using each year's market cap with its XRD variable. This focuses our intensity measurement for a company in the market cap of the current year instead of looking at the intensity of each given year and adding those together. We also agreed to remove the top 100 firms, and the bottom 10% of

firms by market cap for each year. This bottom 10% is a change from the previous iteration where we used a cutoff of \$100 million. We updated to this because as we go farther back in time, the static \$100 million cutoff starts to remove a substantial percentage of companies. Rather than allow this, we now remove a percentage, leaving us with more usable companies. Finally, we decided to keep pharmaceuticals excluded to avoid adding a new variable and make arriving at similar results more likely. We also decided to reconstitute each July and rebalance monthly.

The results with these updates made can be seen in Appendix B. We see similar trends to what we saw before, with the 1980-2000 time frame showing much more significant results than the 2001-2022 time frame. We also see equal weighted portfolios showing higher alphas and greater significance than the value weighted portfolios, which makes it likely that small cap stocks are driving the alphas that we are seeing. When assessing the Fama-French betas that we are seeing for the long-short portfolios, we see the market beta low in all cases. The size factor is consistently positive, with a magnitude between 0.5 and 1 depending on the time period and weighting. The HML factor is near zero for the equal weighted Q5-Q1 portfolios, and slightly positive for the value weighted. The Q5-Zero portfolios, the HML factor is negative, with equal weighted portfolios having more negative betas than value weight. This means that our Q5-Zero portfolios are most likely long on growth stocks and short on value stocks, while the Q5-Q1 portfolios are more balanced.

Our reconciliation efforts have affected the numbers of the outputs that we are seeing, but luckily the trends that we saw in our initial analysis seem to be holding. With our numbers more closely aligned with the other groups, we gain higher confidence that we can trust the numbers coming out of our analysis, and that any conclusions we draw from this data are being made using corroborated and validated results.

Appendix A – Pre Reconciliation Results

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CAPM Results (EW) - 1980-2000 - All Firms								
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	Alpha (%)	Alpha t	Alpha p	Sharpe	Beta (Mkt)	R2		
Q5-Q1	2.0422	4.9174	0.0000	1.3172	-0.0312	0.0006		
Q5-Zero	1.4245	2.9785	0.0033	0.9216	0.2828	0.0375		
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FF3 Results (EW) - 1980-2000 - All Firms								
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	Alpha (%)	Alpha t	Alpha p	Sharpe	Beta (Mkt)	Beta (SMB)	Beta (HML)	R2
Q5-Q1	2.0166	6.0823	0.0000	1.3172	-0.0765	1.0375	0.2028	0.4091
Q5-Zero	1.7259	4.9788	0.0000	0.9216	0.0308	1.0999	-0.3333	0.5309
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CAPM Results (EW) - 2001-2022 - All Firms								
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	Alpha (%)	Alpha t	Alpha p	Sharpe	Beta (Mkt)	R2		
Q5-Q1	0.7948	2.2211	0.0276	0.6898	0.3150	0.0854		
Q5-Zero	0.4203	1.1301	0.2599	0.4152	0.3106	0.0775		
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FF3 Results (EW) - 2001-2022 - All Firms								
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	Alpha (%)	Alpha t	Alpha p	Sharpe	Beta (Mkt)	Beta (SMB)	Beta (HML)	R2
Q5-Q1	0.7589	2.1966	0.0293	0.6898	0.2139	0.5673	0.0037	0.1613
Q5-Zero	0.5291	1.6174	0.1075	0.4152	0.2345	0.5379	-0.6290	0.2980

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CAPM Results (VW) - 1980-2000 - All Firms

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	Alpha (%)	Alpha t	Alpha p	Sharpe	Beta (Mkt)	R2
Q5-Q1	0.3785	1.0304	0.3043	0.2659	-0.0226	0.0004
Q5-Zero	0.3939	1.1801	0.2396	0.4262	0.1685	0.0276

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FF3 Results (VW) - 1980-2000 - All Firms

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	Alpha (%)	Alpha t	Alpha p	Sharpe	Beta (Mkt)	Beta (SMB)	Beta (HML)	R2
Q5-Q1	0.0721	0.2080	0.8355	0.2659	0.1296	0.6453	0.6108	0.1736
Q5-Zero	0.4690	1.4986	0.1359	0.4262	0.0922	0.5027	-0.0478	0.2070

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CAPM Results (VW) - 2001-2022 - All Firms

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	Alpha (%)	Alpha t	Alpha p	Sharpe	Beta (Mkt)	R2
Q5-Q1	0.1412	0.3625	0.7174	0.2920	0.5295	0.1821
Q5-Zero	-0.1059	-0.3280	0.7433	0.1829	0.5727	0.2749

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FF3 Results (VW) - 2001-2022 - All Firms

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	Alpha (%)	Alpha t	Alpha p	Sharpe	Beta (Mkt)	Beta (SMB)	Beta (HML)	R2
Q5-Q1	-0.0575	-0.1719	0.8637	0.2920	0.3820	0.7077	0.6866	0.4064
Q5-Zero	-0.1954	-0.6414	0.5221	0.1829	0.4672	0.5495	0.2460	0.3650

CAPM Results (EW) – 1980-2000 – Filtered (ME>\$100M, Ex-Top100)								
	Alpha (%)	Alpha t	Alpha p	Sharpe	Beta (Mkt)	R2		
Q5-Q1	1.1273	2.5524	0.0116	0.7276	0.0719	0.0029		
Q5-Zero	0.7274	1.3451	0.1804	0.5121	0.3953	0.0562		
FF3 Results (EW) – 1980-2000 – Filtered (ME>\$100M, Ex-Top100)								
	Alpha (%)	Alpha t	Alpha p	Sharpe	Beta (Mkt)	Beta (SMB)	Beta (HML)	R2
Q5-Q1	1.1325	3.0499	0.0027	0.7276	0.0100	0.9958	0.1449	0.3462
Q5-Zero	1.1630	3.0175	0.0029	0.5121	0.0553	1.1790	-0.5451	0.5552
CAPM Results (EW) – 2001-2022 – Filtered (ME>\$100M, Ex-Top100)								
	Alpha (%)	Alpha t	Alpha p	Sharpe	Beta (Mkt)	R2		
Q5-Q1	0.6038	1.7794	0.0768	0.5724	0.2720	0.0718		
Q5-Zero	0.3361	0.9074	0.3654	0.3636	0.3194	0.0822		
FF3 Results (EW) – 2001-2022 – Filtered (ME>\$100M, Ex-Top100)								
	Alpha (%)	Alpha t	Alpha p	Sharpe	Beta (Mkt)	Beta (SMB)	Beta (HML)	R2
Q5-Q1	0.5584	1.6980	0.0912	0.5724	0.1790	0.5114	0.0609	0.1424
Q5-Zero	0.4394	1.3341	0.1839	0.3636	0.2449	0.5240	-0.6006	0.2863

CAPM Results (VW) – 1980-2000 – Filtered (ME>\$100M, Ex-Top100)								
	Alpha (%)	Alpha t	Alpha p	Sharpe	Beta (Mkt)	R2		
Q5-Q1	0.6884	1.8715	0.0630	0.4998	-0.0131	0.0001		
Q5-Zero	0.5770	1.2904	0.1987	0.4895	0.3064	0.0497		
FF3 Results (VW) – 1980-2000 – Filtered (ME>\$100M, Ex-Top100)								
	Alpha (%)	Alpha t	Alpha p	Sharpe	Beta (Mkt)	Beta (SMB)	Beta (HML)	R2
Q5-Q1	0.5653	1.6688	0.0970	0.4998	0.0215	0.7094	0.3148	0.2134
Q5-Zero	0.8748	2.5617	0.0113	0.4895	0.0658	0.9446	-0.3513	0.4858
CAPM Results (VW) – 2001-2022 – Filtered (ME>\$100M, Ex-Top100)								
	Alpha (%)	Alpha t	Alpha p	Sharpe	Beta (Mkt)	R2		
Q5-Q1	0.4816	1.2373	0.2176	0.4599	0.3859	0.1058		
Q5-Zero	0.3092	0.8312	0.4070	0.4026	0.5105	0.1849		
FF3 Results (VW) – 2001-2022 – Filtered (ME>\$100M, Ex-Top100)								
	Alpha (%)	Alpha t	Alpha p	Sharpe	Beta (Mkt)	Beta (SMB)	Beta (HML)	R2
Q5-Q1	0.3304	0.9110	0.3635	0.4599	0.3011	0.3764	0.5669	0.2365
Q5-Zero	0.2955	0.8016	0.4239	0.4026	0.4453	0.3739	-0.0413	0.2130

Appendix B – Post Reconciliation Results

Portfolio	Time Period	Weight Type	Mean Return (%)	Std Dev (%)	Sharpe Ratio	CAPM Alpha (%)	CAPM t-si	CAPM B ₁	CAPM R	FF3 Alpha (%)	FF3 t-si	FF3 Mkt B ₁	FF3 SMB B ₁	FF3 HML B ₁	FF3 R
Zero R&D	1980-2022 (Full)	EW	0.668	5.932	0.39	-0.11	-0.673	1.137	0.735	-0.282	-2.653	1.091	0.735	0.497	0.891
Q1 (Low)	1980-2022 (Full)	EW	0.364	6.605	0.191	-0.499	-2.714	1.261	0.73	-0.434	-3.547	1.113	0.809	-0.148	0.883
Q5 (High)	1980-2022 (Full)	EW	1.736	9.326	0.645	0.732	2.06	1.467	0.495	0.781	3.154	1.208	1.576	-0.076	0.76
High-Low	1980-2022 (Full)	EW	1.372	4.876	0.975	1.231	4.795	0.206	0.036	1.215	5.312	0.095	0.766	0.072	0.249
High-Zero	1980-2022 (Full)	EW	1.068	5.724	0.646	0.842	2.84	0.33	0.067	1.063	4.514	0.118	0.841	-0.573	0.423
Zero R&D	1980-2000 (Early)	EW	0.335	4.672	0.248	-0.373	-1.856	0.921	0.695	-0.575	-3.928	1.01	0.615	0.444	0.85
Q1 (Low)	1980-2000 (Early)	EW	0.024	6.316	0.013	-0.92	-3.286	1.228	0.676	-0.789	-4.361	1.093	0.759	-0.102	0.875
Q5 (High)	1980-2000 (Early)	EW	1.772	9.006	0.682	0.786	1.402	1.282	0.362	0.908	2.637	1.093	1.675	0.064	0.776
High-Low	1980-2000 (Early)	EW	1.747	4.975	1.217	1.706	4.404	0.054	0.002	1.696	5.311	0	0.916	0.166	0.369
High-Zero	1980-2000 (Early)	EW	1.437	6.208	0.802	1.159	2.471	0.361	0.06	1.483	4.376	0.083	1.06	-0.38	0.544
Zero R&D	2001-2022 (Late)	EW	0.974	6.889	0.49	0.185	0.785	1.301	0.787	0.027	0.186	1.123	0.897	0.433	0.923
Q1 (Low)	2001-2022 (Late)	EW	0.676	6.861	0.341	-0.106	-0.442	1.289	0.779	-0.119	-0.731	1.132	0.916	-0.221	0.899
Q5 (High)	2001-2022 (Late)	EW	1.703	9.635	0.612	0.729	1.64	1.606	0.613	0.679	1.917	1.35	1.461	-0.225	0.758
High-Low	2001-2022 (Late)	EW	1.027	4.772	0.745	0.835	2.484	0.317	0.097	0.798	2.464	0.218	0.545	-0.004	0.175
High-Zero	2001-2022 (Late)	EW	0.729	5.233	0.482	0.544	1.457	0.305	0.075	0.652	2.015	0.227	0.564	-0.658	0.315

Zero R&D	1980-2022 (Full)	VW	0.682	4.737	0.499	0.003	0.038	0.991	0.876	-0.104	-1.401	1.005	0.183	0.301	0.916
Q1 (Low)	1980-2022 (Full)	VW	0.689	5.935	0.402	-0.13	-0.947	1.197	0.814	0.01	0.101	1.082	0.403	-0.368	0.911
Q5 (High)	1980-2022 (Full)	VW	1.414	8.051	0.608	0.432	1.658	1.435	0.636	0.384	1.838	1.299	1.015	0.169	0.771
High-Low	1980-2022 (Full)	VW	0.725	4.922	0.51	0.562	2.181	0.238	0.047	0.374	1.603	0.217	0.612	0.537	0.234
High-Zero	1980-2022 (Full)	VW	0.732	5.596	0.453	0.428	1.527	0.444	0.126	0.488	1.992	0.295	0.832	-0.132	0.347
Zero R&D	1980-2000 (Early)	VW	0.562	4.153	0.469	-0.143	-1.239	0.917	0.873	-0.302	-2.787	1.013	0.1	0.288	0.896
Q1 (Low)	1980-2000 (Early)	VW	0.746	6.118	0.422	-0.235	-1.049	1.276	0.779	0.062	0.435	1.057	0.418	-0.44	0.917
Q5 (High)	1980-2000 (Early)	VW	1.441	7.233	0.69	0.494	1.263	1.231	0.519	0.633	2.35	1.073	1.042	-0.068	0.788
High-Low	1980-2000 (Early)	VW	0.695	4.267	0.564	0.729	2.195	-0.045	0.002	0.571	1.845	0.016	0.623	0.372	0.194
High-Zero	1980-2000 (Early)	VW	0.879	5.737	0.531	0.637	1.464	0.314	0.054	0.935	2.874	0.06	0.941	-0.356	0.508
Zero R&D	2001-2022 (Late)	VW	0.791	5.225	0.525	0.156	1.199	1.048	0.887	0.071	0.726	0.979	0.322	0.281	0.937
Q1 (Low)	2001-2022 (Late)	VW	0.637	5.778	0.382	-0.054	-0.329	1.138	0.856	-0.005	-0.036	1.091	0.325	-0.318	0.908
Q5 (High)	2001-2022 (Late)	VW	1.389	8.756	0.55	0.426	1.253	1.588	0.725	0.323	1.044	1.44	0.774	0.225	0.777
High-Low	2001-2022 (Late)	VW	0.753	5.467	0.477	0.48	1.284	0.45	0.149	0.327	0.947	0.349	0.449	0.543	0.286
High-Zero	2001-2022 (Late)	VW	0.598	5.474	0.378	0.27	0.751	0.54	0.215	0.252	0.713	0.46	0.452	-0.056	0.257